Pawan Upadhyay
Independent Scientist | Theoretical Physics & Interdisciplinary Researcher | Economic Science,Soul Science & Biblical Studies

India

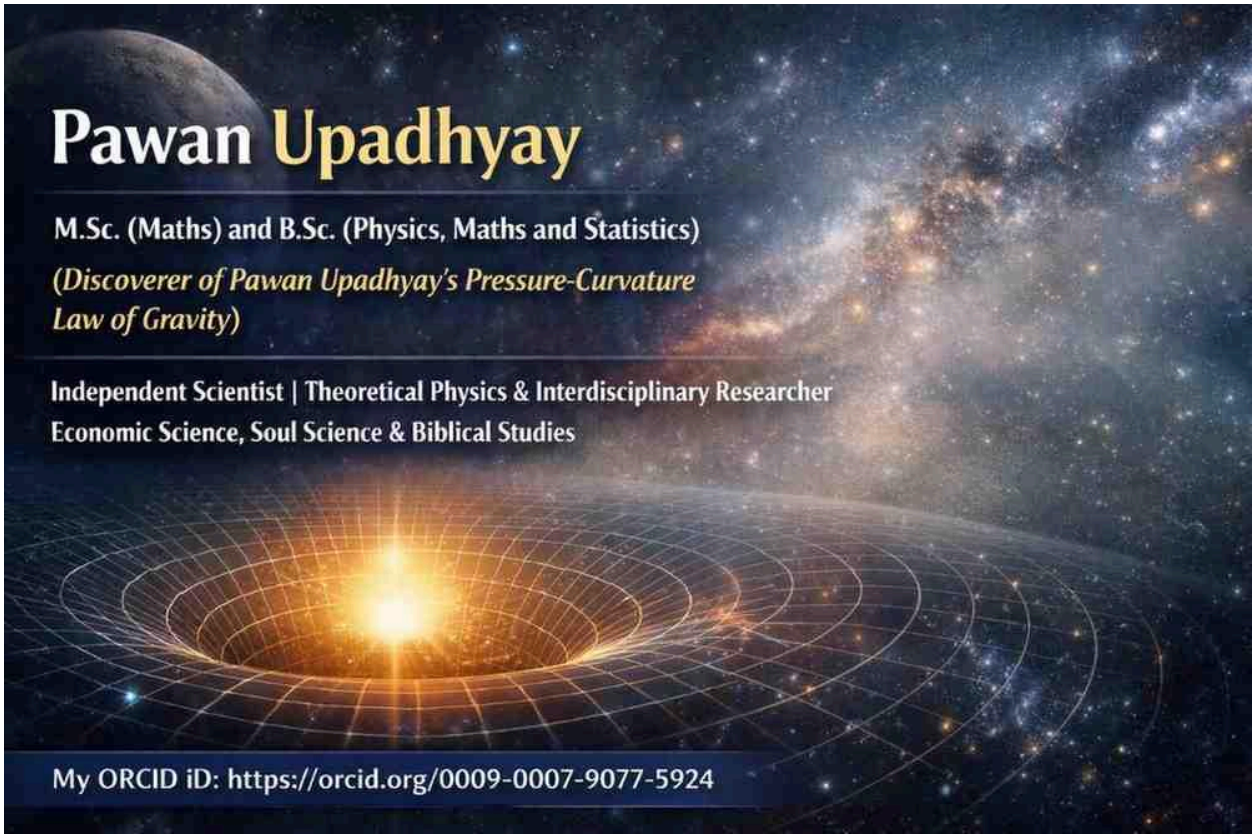
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https://sites.google.com/view/discoveriesbypawanupadhyay/resume



Educational Qualifications

Master of Science (M.Sc.) – Mathematics
Year of Completion: **2009**
University: **Veer Bahadur Singh Purvanchal University, Jaunpur**
Affiliated College: **Harish Chandra Post Graduate College, Varanasi**

Bachelor of Science (B.Sc.) – Mathematics, Physics & Statistics
Year of Completion: **2007**
University: **Veer Bahadur Singh Purvanchal University, Jaunpur**
Affiliated College: **Harish Chandra Post Graduate College, Varanasi**



Professional Summary

Independent scientist and interdisciplinary researcher working at the intersection of Physics, Economic Science, Soul Science, and Biblical Studies. I develop original theoretical frameworks with a focus on fundamental physical laws, economic stability theory, and philosophical questions about consciousness and spiritual reality. I value intellectual freedom, open inquiry, and global accessibility of research.

Professional Status

- Independent ‘Scientist and Researcher’
- Not affiliated with a formal academic institution; research conducted independently with full commitment to academic rigor, open science, and reproducibility.

Scientific Positioning

My work does not propose a replacement for General Relativity, but rather offers a physical interpretation and unifying framework that clarifies the causal role of pressure, energy density, and curvature across gravitational, cosmological, and thermodynamic phenomena.

Primary Scientific Contribution

Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)

Core Principle:

Mass–energy produces pressure, mass-energy bends spacetime by its pressure, pressure generates curvature, and curvature governs motion.

This law provides a mechanical foundation for gravity, long sought since Einstein, by identifying pressure as the missing physical mechanism behind spacetime geometry.

Unified Phenomena through PPC Law:

- Gravity and spacetime curvature
- Time dilation as a pressure-dependent effect
- Entropy evolution in curved spacetime
- Centripetal and centrifugal forces via pressure gradients
- Centripetal force as combination force and centrifugal force as combination force
- Field force and surface force dynamics
- Pressure – curvature dynamics
- Gravitational pressure waves (alternative interpretation of gravitational waves)
- Microgravity
- Role of equation of state parameter in PPC gravity
- Relationship between Energy density and Pressure
- Role of 'field force and surface force'
- 'Field force and surface force' as forces of pressure
- Predictions of the PPC Law
- Interstellar object dynamics
- Cosmic expansion driven by large-scale pressure gradients
- Galactic and planetary binding
- Black hole formation as extreme pressure–curvature collapse
- Event horizons as pressure thresholds
- Wormholes as pressure-supported spacetime tunnels
- Multiverse interpretation via separated pressure domains
- Gravitational Pressure Fields
- Orbital stability and Orbital dynamics
- Light Pressure and Light as Curvature guided wave
- Creation of new 'stars and planets'
- Divergence theorem
- Dark matter and Dark Energy
- Accuracy of PPC Law of Gravity as per modern observation of cosmic acceleration
- Gravitational Lens
- Predicted phenomena as per PPC Law

- Modification of Hawking radiation
- Cosmic acceleration
- Anisotropic stress energy tensor
- Kepler’s Laws in weak field regime
- Energy conservation Law
- Energy density in energy momentum relation
- Velocity in weak field regime and strong field regime
- Vector structure of Forces
- Gravitational Pressure Field, Gravitational field and Pressure Gradient field in weak field limit
- Information conservation in PPC Law of Gravity
- Non-True Singular Black holes in PPC Law of Gravity
- Gravitational pressure field in Strong field regime
- Energy density and Frequency density
- Energy conservation law and Acceleration
- Kinetic energy density and Kinetic pressure
- Speed of Light and Light Pressure Effect
- Observational 'evidence and consistency' of PPC Law of Gravity
- No static black holes principle in PPC Law of Gravity
- Weightlessness, Free Fall and Air Pressure Drag
- Relationship between Gravitational Pressure Field, Pressure Gradient Field and Gravitational Field
- Four acceleration, Four field force density and Four velocity in Strong field Regime
- Momentum density and Four field force density
- Mass density and Energy density equivalence
- Accuracy of Pawan Upadhyay’s Pressure Curvature Law of Gravity in Predicting Mercury’s Perihelion Precession
- Universal Geodesic Motion Principle (UGMP)
- Wavelength density and frequency density
- Wavelength density and momentum density
- Wavelength density and Energy density
- Potential Energy density and mass density
- Accelerated cosmic expansion

Key Contributions & Distinctions

- Proposed a unified physical framework (PPC Law) identifying pressure as the causal mechanism of spacetime curvature, providing physical intuition alongside General Relativity.
- Introduced the dual-force structure of gravity via field force (pressure gradient) and surface force (pressure acting on area).
- Reinterpreted gravitational waves as pressure–curvature waves within an equivalent relativistic framework.

- Provided pressure-based interpretations of time dilation, entropy, wormholes, microgravity, blackholes, galaxies, multiverse, cosmic expansion, dark matter, dark energy, light pressure and orbital dynamics.
- Creation of new equations
- Densities based formulation

Research Methodology & Approach

- Development of original theoretical models grounded in established physical principles.
- Consistent alignment with experimentally verified results of General Relativity.
- Use of mathematical reasoning, conceptual analysis, and physical interpretation.
- Emphasis on causal clarity, physical intuition, and unification of concepts.
- Open peer engagement through public repositories and open-access platforms.

Scientific Skills

- Theoretical & Mathematical
- Theoretical Physics
- Mathematical Physics
- General Relativity (interpretation-based analysis)
- Relativistic Cosmology
- Gravitational Theory
- Classical Mechanics
- Fluid Mechanics
- Thermodynamics

Advanced Mathematical & Analytical Skills

- Tensor Calculus
- Differential Geometry
- Vector and Tensor Analysis
- Dimensional Analysis and Units Consistency
- Equation-of-State Analysis
- Mathematical Modeling of Physical Systems
- Gravity & Cosmology Analysis
- Stress–Energy Tensor Interpretation
- Energy Density and Pressure Modeling
- Gravitational Pressure and Force Density Analysis
- Field Force and Surface Force Modeling
- Cosmological Acceleration Analysis
- Dark Energy and Dark Matter Interpretation
- Physical Modeling & Conceptual Frameworks
- Pressure–Curvature Modeling
- Force Density and Surface Force Formulation

- Continuum Mechanics Interpretation
- Physical Interpretation of Einstein's Equations
- Causal Analysis of Spacetime Dynamics
- Modification of Einstein's Equations

Research & Scientific Practice

- Independent Scientific Research
- Original Theory Development
- Scientific Writing and Documentation
- Open-Access Research Publishing (OSF, Archive.org)
- Conceptual Unification of Physical Theories

Research Areas

- Theoretical Gravity and Cosmology
- Pressure-Induced Spacetime Geometry
- Black hole and Wormhole Physics
- Galactic and Planetary Binding Mechanisms
- Microgravity and Interstellar Objects
- Energy Density and Equation-of-State in Gravity
- Entropy and Thermodynamics of Spacetime
- Gravitational Pressure Waves
- Gravitational Pressure fields
- Pressure Forces

Interdisciplinary Research :

Economic Science

- Theory of cash inflow and cash outflow
- Fund creation mechanisms
- Economic stability models
- Three Fund Stabilization Model
- Future Cash inflows theories
- Economic Science based on 'Cash inflows and Cash outflows' theories with mathematical equations
- Solutions of the Economic Problems
- Mathematical framework development for theoretical economic equations
- Equation-based modeling of economic systems
- Difference between investment and purchase
- Consequences of currency manipulation

Soul Science & Consciousness Studies

- Philosophical investigation of soul–body interaction
- Consciousness beyond physical reduction
- Role of Soul in ‘Human Body and Plant Body'
- Biological explanation
- Explanation of Soul Science in the language of Biology.
- Soul as Energy and Soul as internal force
- Study of consciousness

Biblical Philosophy

- ‘Resurrection and Reincarnation’ studies in Christianity
- Distinction between soul, spirit, and physical body
- Difference between Hell and Satan
- New Testament Studies
- Greek words studies
- Old Testament Studies

Biblical Studies – Research Domains

- Christology (death, descent to the dead, resurrection, ascension)
- Afterlife theology (Sheol, Abraham’s Bosom, Paradise, Heaven)
- Salvation history in Old and New Testament theology
- Biblical cosmology and heavenly realms
- Second Temple Jewish and apocalyptic literature
- Canonical and non-canonical textual comparison

Research Methodology (Biblical Studies)

- Historical-grammatical exegesis of biblical texts
- Chronological reconstruction of New Testament events
- Thematic synthesis across multiple scriptural sources
- Distinction between canonical doctrine, theological inference, and interpretive models
- Comparative analysis of biblical and apocalyptic cosmologies

Ongoing & Future Research Projects

- Comprehensive Christology research series
- Afterlife realms in Biblical theology
- Heavenly hierarchies and cosmology from Genesis to Revelation
- Comparative study of Old Testament, New Testament, and Enochic traditions

Research Ethics & Scholarly Practice

- Commitment to academic integrity and responsible citation
- Clear separation of research analysis from doctrinal assertion
- Methodologically transparent theological inquiry
- Respect for canonical boundaries and historical context

Publications & Open Access

- Research papers published with DOIs on Open Science Framework (OSF)
- Long-term preservation and public access via Archive.org

Archive Profile:
https://archive.org/details/@pawan_upadhyay

All research documents are openly available for academic and public use.

Open Science Statement

I am committed to open-access knowledge, transparency, and independent scientific inquiry. All my works are freely accessible to ensure global participation and long-term preservation of original research.

Research Vision

To advance a physically intuitive understanding of fundamental laws of nature, encourage interdisciplinary dialogue, and make original scientific thought globally accessible through open research.

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Signature
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https://about.me/pawanupadhyay
https://doi.org/10.5281/zenodo.21475601
https://osf.io/b7ygf/
https://github.com/pawanupadhyay2025